(2)

(Please write your Enrollment Number) Enrollment No. 194

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B.Tech (CSE AI) , Semester:5
(December, 2025)

Subject Code: BAI 301	Subject: Machine Learning
Time :3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8=20)	CO Mapping
	a) What role do evaluation metrics play in assessing model performance?		CO1
	b) What are *Data Objects* and *Attributes*?		CO1
	c) How does the choice of kernel affect model performance in non-linear classification?		CO2
	d) Compare Ridge and LASSO regression in terms of regularization and feature selection.		CO2
	e) How can PCA help improve clustering performance?		CO3
	f) How does reducing the number of features impact model accuracy and computation time?		CO3
	g) Explain the concept of k-fold cross-validation with an example.		CO4
	h) What is the purpose of a *learning curve* in machine learning?		CO4
	UNIT I		CO Mapping
Q2	Discuss the importance of data preprocessing in Machine Learning. Describe the steps involved in Data Cleaning, Integration, and Feature Extraction.	(10)	CO1
Q3	Compare supervised and unsupervised learning with suitable examples.	(10)	CO1
	UNIT II		CO Mapping
Q4	Discuss the working of Support Vector Machines. Explain how kernel functions enable SVMs to handle non-linear classification.	(10)	CO2
Q5	A company wants to predict whether a new employee will leave or stay, based on past data. The dataset is as follows:	(10)	CO2

Employee	Salary	Experience	Leaves Company
E_1	High	Senior	No
E_2	High	Junior	No
E_3	Low	Junior	Yes
E_4	Low	Senior	Yes
E_5	Low	Junior	Yes
E_6	High	Senior	No

You are asked to classify a new employee with:
Salary = Low, Experience = Senior.

Scanned with OKEN Scanner

	UNIT III		CO Mapping
Q6	Two classes of data points are given as follows: | Class	X_1	X_2
---	---	---
C_1	1	2
C_1	2	3
C_1	3	3
C_2	6	5
C_2	7	8
C_2	8	6 | Project all samples from both classes onto the line using Linear Discriminant Analysis.	(10)	CO3
Q7	The following table shows a dataset consisting of five observations and two features: | Observation	X_1	X_2
---	---	---
1	2	0
2	3	1
3	4	0
4	5	2
5	6	3 | *Transform the original data into the new coordinate system Using Principal Component Analysis.*	(10)	CO3
	UNIT IV		**CO Mapping**
Q8	*Discuss the concept of statistical hypothesis testing and its use in comparing machine learning models.*	(10)	CO4
Q9	***Explain the concept of Ensemble Classification.*** *Describe how ensemble methods like Random Forest and AdaBoost achieve higher accuracy through diversity.*	(10)	CO4

Scanned with OKEN Scanner